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Multi-granularity awareness via cross fusion for few-shot learning

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ABSTRACT

Many existing few-shot learning methods based on a single-granularity structure have achieved promising results, but they insufficiently exploit the latent information within granularity structures. In addition, these methods still struggle with unclear class distinctions and scattered intra-class feature spaces, posing significant challenges for few-shot learning. To address these issues, we propose a multi-granularity awareness-guided cross-fusion (MACF) method for few-shot image classification. MACF enhances inter-class separation while promoting intra-class cohesion. Specifically, we introduce multi-granularity awareness to construct both coarse- and fine-grained representations, strengthening class affiliations and improving feature discriminability across classes. Moreover, we utilize local descriptors as features to develop a cross-fusion strategy that generates distinguishable pseudo-features, enriching intra-class feature diversity and mitigating the impaired generalization ability for unseen classes. Furthermore, we establish a reliability assessment metric for synthetic features to ensure the generation of representative features while reducing the impact of outlier perturbations. Extensive experiments on six few-shot benchmark datasets demonstrate the effectiveness of the proposed MACF method. The basic data and code can be accessed via the link: https://github.com/woodszp/MACF.

1. Introduction

Humans acquire knowledge from only a few examples and can recognize new classes accordingly [1]. In contrast, traditional machine learning and deep learning methods require a large number of labeled samples to train models, limiting their applicability. This limitation is particularly concerning because obtaining extensive labeled data in complex environments such as medical image classification during the early stages of an epidemic is extremely challenging. In such cases, traditional deep learning struggles to learn models that recognize new classes owing to the scarcity of training samples. To address this challenge, few-shot learning (FSL) has emerged as a solution, enabling models to extract latent knowledge from a limited number of labeled samples, much like how humans learn [2].

FSL has garnered significant attention owing to its potential value in artificial intelligence, particularly in tasks such as semantic segmentation, object detection, and image classification [3]. To address the challenges of FSL, researchers focus on developing meta-learning models that extract meta-knowledge from base classes and transfer it to unseen classes to facilitate learning [4]. These meta-learning models are generally categorized into optimization-based and metric-based approaches. Optimization-based methods aim to learn an optimizer capable of rapidly updating model parameters to adapt to unseen tasks. In contrast, metric-based methods [5] seek to establish a relationship that effectively measures the similarity between query samples and support samples.

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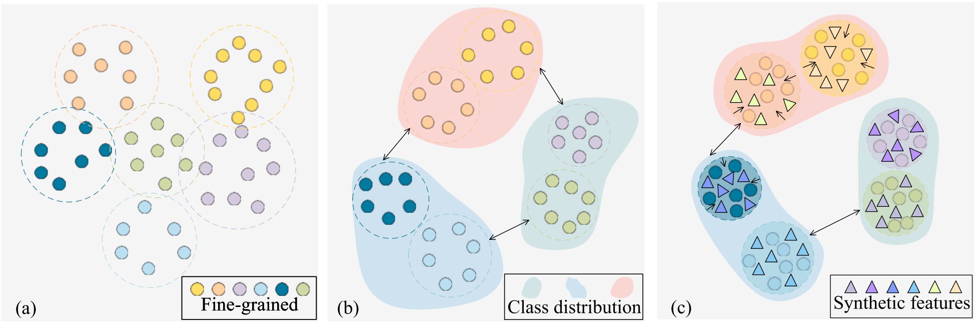

Fig. 1. The main idea of our method. Different colors denote features from different classes, and dashed circles with different colors represent the class feature space extent. (a) The single-granularity structure method is difficult to distinguish inter-classes feature space, and the intra-class feature space is scattered, affecting classification performance. (b) We leverage multi-granularity awareness to construct coarse- and fine-grained class representations to guide similar samples to approach each other, and heterogeneous samples are far from each other. (c) We use Gaussian cross-fusion to cross-fuse fine-grained local description features to generate representative synthetic features, enriching the diversity of the intra-class feature space.

Recently, some researchers have explored metric-based learning methods that leverage granular information and data augmentation techniques to mitigate the data scarcity problem and achieve effective results [6]. For example, Zhang et al. [7] designed a cross-referencing strategy on the single fine-grained level to reduce the impact of the background. Additionally, Singh et al. [8] expanded the training set by rotating images at different angles within a single granularity. Despite their promising results, these approaches still struggle with unclear distinctions among fine-grained classes and scattered intra-class distributions, as shown in Fig. 1(a). These challenges significantly impact the effectiveness of the FSL classification task.

For inter-class discrimination, if the distinction between granularities in the feature space is clearer and the internal structure of each granularity is more stable, class differentiation can be further improved [9]. Thus, we focus on the relationships among class granularities and further explore their affiliations, as illustrated in Fig. 1(b). We explore the information transfer between different hidden layers of the network by establishing a multi-granularity awareness module. This approach enriches the extracted features with more detailed texture representations, enhancing inter-class feature distinction.

To enhance intra-class feature diversity, we design an innovative and flexible cross-fusion strategy to generate reliable synthetic features and enrich intra-class granularity modal diversity, as illustrated in Fig. 1(c). This strategy leverages data semantic knowledge to guide the cross-fusion of features, resulting in a more compact and stable intra-class feature space distribution while improving the generalization ability of the model. Furthermore, we propose the multi-granularity awareness-guided cross-fusion (MACF) method for few-shot image classification, which focuses on feature-level inter-class discrimination and intra-class diversity. This approach facilitates class separation while reinforcing feature aggregation within each class.

Specifically, we construct distributional differences in the class feature space using a *multi-granularity awareness* module and employ a *Gaussian cross-fusion* strategy to learn fine-grained regional features, addressing the limited generalization ability of FSL. Our method comprises three key components: the *multi-granularity awareness* module, the *Gaussian cross-fusion* strategy, and *reliability assessment*, which iteratively enhance each other during the learning process. In particular, the *multi-granularity awareness* module enriches the structural information among classes and further uncovers inter-class associations. Unlike traditional single-granularity structures, our multi-granularity framework incorporates both coarse- and fine-grained class affiliations, facilitating more effective information transmission. We employ the *Gaussian cross-fusion* strategy to merge fine-grained local features from support and query samples, generating synthetic features that are incorporated into the training process. This approach enhances intra-class feature diversity while reducing spatial distribution differences within each class. In addition, *reliability assessment* evaluates the distributional differences between Gaussian fusion-generated features and original category features. By leveraging these assessments, the method constrains the representation of synthetic features, ensuring a more compact spatial distribution and minimizing inter-class feature discrepancies.

The contributions of this paper are summarized as follows:

- We introduce an innovative multi-granularity awareness structure that leverages rich information across different granularities. This structure addresses the challenge of ambiguous category distinctions, enhances the understanding of class relationships, and facilitates the extraction of more discriminative features.
- We propose a cross-fusion strategy as a plug-and-play solution for synthesizing pseudo-features by integrating fine-grained features from both support and query samples. This approach mitigates the issue of intra-class modal singularity caused by the lack of diversity in similar features, thereby enabling a more comprehensive representation of potential class distributions.
- We conduct extensive experiments on six FSL benchmark datasets. Our proposed method consistently outperforms existing FSL approaches, demonstrating its effectiveness and robustness.

The rest structure of this paper is outlined as follows: Section 2 provides a overview of related work in the field. Section 3 presents the implementation of the MACF algorithm. Section 4 presents the results of our experiments conducted on various datasets. Finally, concluding remarks and prospective directions for subsequent research are articulated in Section 6.

2. Related work

In this section, we provide a brief overview of related work, categorizing it into meta-learning based FSL, multi-granularity FSL, and data augmentation-based FSL.

2.1. Meta-learning based FSL

The rapid advancement and widespread application of meta-learning strategies in deep learning have significantly improved the classification capabilities of FSL across various scenarios. Recently, most existing meta-learning methods can be broadly classified into optimization-based methods [10] and metric-based methods [11].

Optimization-based methods focus on learning an effective initialization of model parameters, enabling rapid adaptation to new tasks with just a few gradient descent steps. Model-agnostic meta-learning (MAML) [10] is a representative approach in this category, designed to quickly update parameters using a meta-learning framework that leverages a second-order optimization strategy. However, using second derivatives in MAML makes the optimization process computationally intensive and cumbersome. To address this issue, Nichol et al. [12] proposed a first-order derivative method, which simplifies the optimization process while maintaining competitive performance.

Metric-based methods aim to measure the similarity between feature embeddings of different classes, often using Euclidean distance or cosine similarity [11]. For example, Kang et al. [13] explored self-correlational representation and cross-correlational attention, leveraging relational patterns both within and between images. Due to their simplicity and efficiency, metric-based methods have gained significant attention in FSL. Representative approaches include matching networks [14], ProtoNet [15], relation networks [16], DN4 [17], and RENet [13].

However, these methods primarily focus on pairwise or local relationships, overlooking the multi-granularity associations both within and across classes. In contrast, our proposed method incorporates a multi-granularity awareness module that not only captures relational patterns across different granularities but also further explores intra-class associations, enhancing feature representation and classification performance.

2.2. Multi-granularity FSL

Multi-granularity FSL leverages the inherent granularity structure of data or the relationships among different granularities to enhance FSL and classification [18]. Granular classification is widely applied in real-world tasks, as it decomposes complex problems into smaller, more manageable sub-tasks [19]. Additionally, it improves classification accuracy by accommodating various levels of granularity. For example, Liu et al. [6] proposed a coarse-to-fine classifier that utilizes class hierarchies as prior knowledge to guide the model in learning different granular representations. Similarly, Zhu et al. [20] investigated contrastive learning techniques to extract more general and transferable features in multi-granularity training scenarios.

Previous multi-granularity methods primarily focus on the semantic hierarchy of data, structuring information in a layered manner based on this hierarchy. However, our proposed method differs significantly in that it integrates the granularity of the structure directly into the feature extraction process. Specifically, our approach establishes a strong correlation between granularity levels and extracted features, enabling the feature extraction network to uncover rich texture information both within and across layers.

2.3. Data augmentation-based FSL

Several data augmentation-based approaches have been proposed for FSL. These methods can be broadly categorized into pixel-level augmentation [21] and feature-level hallucination techniques [22].

Pixel-level augmentation techniques usually generate new image representations by applying transformations such as hallucination, shrinking, the mixing of paired samples, or the random erasing of parts of an image [7]. To achieve this goal, various methods have been proposed, including using an attentive PixelCNN network and MetaGAN [21]. For example, Singh et al. [8] enhanced model training by rotating the center and edge positions of images at different angles to create new augmented samples. **Feature-level hallucination** focuses on modifying image features to enhance model performance. Cao et al. [5] proposed a semantic alignment model to address content misalignment by ensuring feature consistency. Recently, Zhang et al. [22] proposed a feature generation method that mitigates the large distribution gap between generated and real data. Overall, data augmentation-based FSL methods continue to advance through improvements in both pixel-level transformations and feature-level hallucination techniques.

However, these methods overlook the importance of granularity structure in FSL classification tasks. Our approach fully leverages latent granularity knowledge to guide and fuse object and edge positions within fine-grained classes, thereby enriching inter-class diversity. Moreover, we introduce a synthetic feature measurement strategy to enhance the effectiveness of fused features.

3. Method

This section has five parts. We first define the few-shot learning problem. Then, we introduce our proposed algorithm, which consists of a multi-granularity awareness structure, synthesis features via cross-fusion, and reliability assessment. Finally, we summarize the objective of the network.

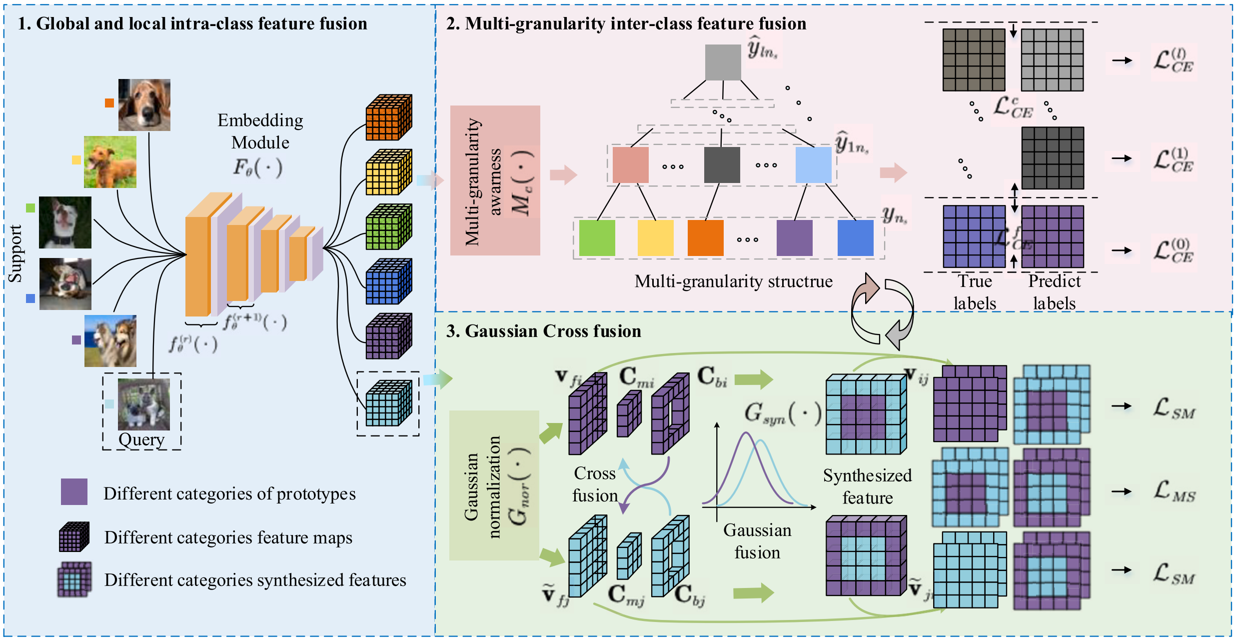

Fig. 2. Illustration of MACF framework.

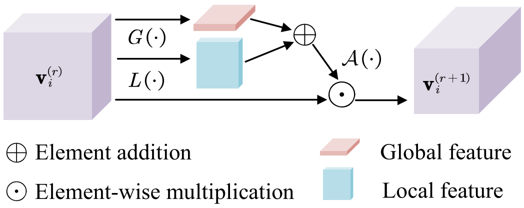

Fig. 3. Illustration of the global and local intra-class feature fusion process.

3.1. *Problem definition*

For FSL, we adopt the episodic training paradigm [7]. In this paper, each N-way K-shot training classification task is denoted as $\mathcal{T}_{train}$. Similarly, the test classification task is denoted as $\mathcal{T}_{test}$. For each task $\mathcal{T}_{train} = \{\mathbf{X}_s, \tilde{\mathbf{X}}_q, \hat{\mathbf{Y}}_c, \mathbf{Y}_f\}$, it contains four parts of data, where $\mathbf{X}_s = (\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_{n_s})$ represents the support samples, n_s is the number of support samples. Matrix $\tilde{\mathbf{X}}_q = (\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \cdots, \tilde{\mathbf{x}}_{n_q})$ denotes the query samples, where n_q is the number of query samples, $\hat{\mathbf{Y}}_c = (\hat{y}_{1n_s}, \hat{y}_{2n_s}, \cdots, \hat{y}_{ln_s})$ represents the corresponding coarse-grained label matrix of support samples, l is granularity layer marker, and $\mathbf{Y}_f = (y_1, y_2, \cdots, y_{n_s})$ denotes the corresponding fine-grained label matrix of support samples. Only datasets with hierarchies have coarse-grained label $\hat{\mathbf{Y}}_c$, and tasks for datasets without hierarchies can be denoted as $\mathcal{T}_{train} = \{\mathbf{X}_s, \tilde{\mathbf{X}}_q, \mathbf{Y}_f\}$.

The proposed multi-granularity awareness module integrates the potential feature information among layers, enriching the feature representation of samples to help reveal different granular representations. We perform global and local feature fusion after each layer of the network to avoid the intra-class distinction caused by the tiny distinction among fine-grained features within the intra-class, as shown in Fig. 2.

3.2. *Global and local intra-class feature fusion*

For a specific task, the way to obtain potential feature information among granular layers is shown in Fig. 3, and the fusion calculation is as follows:

$$\mathbf{v}_i^{(r+1)} = f_\theta^{(r)}(\mathbf{x}_i) = \mathbf{v}_i^{(r)} \odot \mathcal{A}(L(\mathbf{v}_i^{(r)}) \oplus G(\mathbf{v}_i^{(r)})), \tag{1}$$

where $\mathbf{v}_i^{(r)}$ means the i-th feature of the sample obtained from the r-th layer of the embedding network, $\odot$ represents element-wise multiplication, $\mathcal{A}(\cdot)$ acts as attention computation function, $\oplus$ is element addition, $L(\cdot)$ and $G(\cdot)$ represents local and global channel attention computation. Then, we obtain the fused features $\mathbf{v}_i^{(r+1)}$ of the current layer $f_\theta^{(r)}(\cdot)$ and then deliver the fused features to the next layer of the embedding network $f_\theta^{(r+1)}(\cdot)$. Especially, when $r=0$ the $f_\theta^{(0)}(\mathbf{x}_i) \to \mathbf{v}_i^{(1)}$. Finally, the representation of the embedding module $F_\theta(\cdot)$ refined to each layer is $f_\theta^{(r)}(\cdot)$, where r indicates the r-th layer of the network: $F_\theta(\mathbf{x}_i) = f_\theta^{(r+1)}(f_\theta^{(r)}(\mathbf{x}_i))$.

3.3. Multi-granularity inter-class feature fusion

For convenience, we define the granularity other than fine-grained as coarse-grained, due to the granularity has a multi-level representation. As shown in Fig. 2, given only a few samples to extract features, and then the coarse and fine granularity is constructed through the multi-granularity perceptron. Coarse- and fine-grained features are mapped through the multi-granularity awareness $M_c(\cdot)$:

$$M_c(\mathbf{v}_i) = \begin{cases} \mathbf{v}_{ci}, l \geq 1; \\ \mathbf{v}_{fi}, l = 0, \end{cases} \tag{2}$$

where $\mathbf{v}_{ci}$ and $\mathbf{v}_{fi}$ represent the coarse- and fine-grained sample features of the i-th sample, respectively. For the sake of simplicity, we use $\mathbf{v}_{ci}$ to uniformly represent the coarse-grained sample feature for different layers. Inter-class discrimination ensures that the extracted sample features can be correctly classified into the coarse-grained classes they belong to and are sufficiently discriminative to distinguish different classes.

Specifically, we first obtain coarse-grained features $\mathbf{v}_{ci}$ of the i-th sample. Then, we utilize a coarse-grained linear classifier $H_c(\cdot)$ to predict the coarse-grained labels of the samples as follows:

$$\hat{y}_{ci}^* = H_c(\mathbf{v}_{ci}) = \mathbf{v}_{ci}\mathbf{A}_c^\top + b_c, \tag{3}$$

where $\mathbf{A}_c^\top$ is the weight matrix of the coarse-grained classifier, b_c is the bias coefficient of the coarse-grained classifier. We guide inter-class learning among granularities via semantic knowledge of hierarchical data. Similarly, we set the weight matrix of the fine-grained classifier $\mathbf{A}_f^\top$ and set the bias coefficient b_f to get the fine-grained prediction label $y_i^* = \mathbf{v}_{fi}\mathbf{A}_f^\top + b_f$.

Similarity measure. To increase the discrimination between heterogeneous samples, we construct the coarse-grained classification loss of the samples guided by coarse-grained semantic knowledge as follows:

$$\begin{aligned} \mathcal{L}_{CE}^c &= -\frac{1}{l}\sum\nolimits_{k=1}^{l} \mathcal{L}_{CE}^{(k)} \\ &= -\frac{1}{l \cdot n_s}\sum\nolimits_{k=1}^{l}\sum\nolimits_{i=1}^{n_s} \log \mathcal{I}(\hat{y}_{ci}^*, \hat{y}_{ci}) \\ &= -\frac{1}{l \cdot n_s}\sum\nolimits_{k=1}^{l}\sum\nolimits_{i=1}^{n_s} \log \hat{p}_{ci}, \end{aligned} \tag{4}$$

where $\mathcal{L}_{CE}^c$ is the sum of the coarse-grained cross-entropy loss, which is calculated by different granularity layers loss $\mathcal{L}_{CE}^{(l)}$, l denotes coarse-grained layers, n_s is the number of support samples in the current. $\hat{p}_{ci}$ represents the probability that the sample correctly distinguishes the class to which the coarse-grained belongs, $\mathcal{I}(\cdot)$ is a discriminant function that determines whether the predicted label $\hat{y}_{ci}^*$ of the sample is consistent with the true label $\hat{y}_{ci}$. The coarse-grained class discrimination promotes the fine-grained class space contraction to a certain extent. On the one hand, to further reduce the intra-class differences of fine-grained classes, and on the other hand, to provide high-quality granular features to prepare for Gaussian fusion. Therefore, we take the cross-entropy calculation method to correct the prediction results of the fine-grained linear classifier as follows:

$$\mathcal{L}_{CE}^f = \mathcal{L}_{CE}^{(0)} = -\frac{1}{n_s}\sum\nolimits_{i=1}^{n_s} \log p_i, \tag{5}$$

where $\mathcal{L}_{CE}^f$ is the fine-grained cross-entropy loss, which is expressed as the cross-entropy loss $\mathcal{L}_{CE}^{(0)}$ $p_i = \mathcal{I}(y_i^*, y_i)$ represents the probability that the sample correctly distinguishes the class to which the fine-grained belongs.

3.4. Gaussian cross fusion

Traditional FSL adopts the similarity measure between the support and the query samples to achieve excellent performance, whose details are as follows:

$$\mathcal{L}_{Mb} = -\log \frac{\exp(sim(\mathbf{v}_{fi}, \tilde{\mathbf{v}}_{fj}))}{\sum_{i=1}^{n_s}\sum_{j=1}^{n_q} \exp(sim(\mathbf{v}_{fi}, \tilde{\mathbf{v}}_{fj}))}, \tag{6}$$

where $sim(\cdot)$ is the cosine similarity function, $\mathbf{v}_{fi}$ is the fine-grained feature of the i-th support sample, and $\tilde{\mathbf{v}}_{fj}$ is the fine-grained feature of the j-th query sample. Unlike the existing FSL methods investigating the data augmentation strategy, we suggest the cross-fusion of the center and edge position of the image descriptor to generate pseudo-features to assist training instead of leveraging the generator.

Given one or a few training samples, let the position of the image center point be (t_x, t_y), and let t_w be half the width of the mask. As shown in Fig. 4, we set $t_w = \frac{H}{4}$ to obtain the sample center position feature. The detailed calculation is as follows:

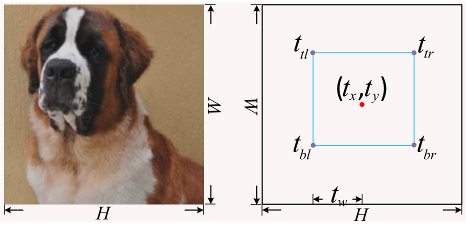

Fig. 4. Illustration of the Gaussian fusion module selecting center and edge positions of the sample and calculating each pixel.

$$\begin{aligned} t_{tl} &= (t_x - t_w, t_y + t_w), \\ t_{tr} &= (t_x + t_w, t_y + t_w), \\ t_{bl} &= (t_x - t_w, t_y - t_w), \\ t_{br} &= (t_x + t_w, t_y - t_w), \end{aligned} \tag{7}$$

where t_{tl}, t_{tr}, t_{bl}, and t_{br} represent the pixel coordinates of the top left, top right, bottom left, and bottom right positions of the mask, respectively.

Gaussian cross fusion process mainly consists of two steps:

- We perform Gaussian normalization $G_{nor}(\cdot)$ on the feature $\mathbf{v}_{fi}$, and split the features map $\mathbf{v}_{fi}$ into the center position descriptor $\mathbf{C}_{mi}$ and edge position descriptor $\mathbf{C}_{bi}$ in the spatial dimension of the features.
- We utilize Gaussian fusion $G_{syn}(\cdot)$ to cross-fuse the center location descriptor features of the support samples $\mathbf{C}_{mi}$ and the edge location descriptor features of the query samples $\tilde{\mathbf{C}}_{bj}$, respectively.

Considering the differences between query and support sample features, direct fusion without Gaussian normalization limits the learning ability. First, we suggest applying Gaussian normalization $G_{nor}(\cdot)$ on the fused features to improve the correlation between the center and edge position features.

To obtain the feature descriptors of the center and edge locations of the samples, we define a mask function $M_{mi}(t_x, t_y, t_w)$ that helps in splitting the feature map $\mathbf{v}_{fi}$. The detailed calculation is as follows:

$$M_{mi}(t_x, t_y, t_w) = \begin{bmatrix} t_{tl} & t_{tr} \\ t_{bl} & t_{br} \end{bmatrix} = \begin{bmatrix} (t_x - t_w, t_y + t_w) & (t_x + t_w, t_y + t_w) \\ (t_x - t_w, t_y - t_w) & (t_x + t_w, t_y - t_w) \end{bmatrix}. \tag{8}$$

Using this mask function, we can derive the feature descriptors of the center and edge locations as follows:

$$\begin{cases} \mathbf{C}_{mi} = \mathbf{v}_{fi} \odot M_{mi}(t_x, t_y, t_w); \\ \mathbf{C}_{bi} = \mathbf{v}_{fi} - \mathbf{C}_{mi}, \end{cases} \tag{9}$$

where symbol $\odot$ represents element-wise multiplication, $\mathbf{C}_{mi} \in \mathbb{R}^{C \times \frac{H}{2} \times \frac{W}{2}}$ and $\mathbf{C}_{bi} \in \mathbb{R}^{C \times \frac{\sqrt{3}H}{2} \times \frac{\sqrt{3}W}{2}}$. Here, C, H, and W represent the number of channels, height and width of the feature, respectively. Similarly, we obtain the features of the center position $\tilde{\mathbf{C}}_{mi}$ and edge position $\tilde{\mathbf{C}}_{bi}$ of the query set according to the above calculation.

Finally, to maintain the robustness of the network, we synthesized the features of the Gaussian normalized center and edge positions in the spatial position of the feature, which aims to remain the measure dimensions consistent between the support and query samples in the end-to-end training as follows:

$$\begin{aligned} \mathfrak{v}_{ij} &= G_{syn}(\Theta, G_{nor}(\mathbf{C}_{mi}), G_{nor}(\tilde{\mathbf{C}}_{bj})); \\ \tilde{\mathfrak{v}}_{ji} &= G_{syn}(\Theta, G_{nor}(\tilde{\mathbf{C}}_{mj}), G_{nor}(\mathbf{C}_{bi})), \end{aligned} \tag{10}$$

where $G_{syn}(\cdot)$ represents the Gaussian fusion function and Θ is the scaling factor. $\mathfrak{v}_{ij}$ is the cross-fusion feature of the center position feature of the i-th support sample and the edge position of the j-th query sample, $\tilde{\mathfrak{v}}_{ji}$ is the cross-fusion feature of the center position feature of the j-th query sample and the edge position of the i-th support sample.

3.5. Reliability assessment

In terms of the distinguishability of the synthetic pseudo-features, to avoid affecting the distribution of the original intra-class feature space since the participation of the synthetic pseudo-features. Thereby, the synthetic pseudo-features maintain a similar distribution to their original center position features as possible. We design a reliability assessment between the synthesized pseudo-features and their central position original sample features as follows:

$$\mathcal{L}_{SM} = \frac{1}{n_s} \frac{1}{n_q} \sum_{i=1}^{n_s} \sum_{j=1}^{n_q} (d(\mathfrak{v}_{ij}, \mathbf{v}_{fi}) + d(\tilde{\mathfrak{v}}_{ji}, \tilde{\mathbf{v}}_{fj})), \tag{11}$$

Table 1
Benchmark datasets description. Symbol "-" indicates not applicable.

Description	tieredImageNet	FC100	CIFAR-FS
structure	Y	Y	Y
depth	3	3	3
coarse train/val/test	20/6/8	12/4/4	20/11/13
fine train/val/test	351/97/160	60/20/20	60/16/20
all fine classes	608	100	100
sample	779,165	60,000	60,000
size	$84 \times 84 \times 3$	$32 \times 32 \times 3$	$32 \times 32 \times 3$
Description	**miniImageNet**	**CUB**	**Cars**
structure	N	N	N
depth	2	2	2
coarse train/val/test	-	-	-
fine train/val/test	64/16/20	100/50/50	100/48/48
all fine classes	100	200	196
sample	60,000	11,788	16,185
size	$84 \times 84 \times 3$	-	$360 \times 240 \times 3$

where $d(\cdot)$ is the distance metric function, n_s and n_q are the numbers of support and query samples, respectively. The intra-class spatial sparsity problem caused by a single intra-class modality can be alleviated by measuring the similarity between synthetic pseudo-features. Since the synthesized pseudo-features act as different modalities of the same class in the original intra-class space. It promotes the intra-class to have enough diversity to represent as many class modalities as possible and avoids the lack of model generalization ability due to modal singularity. The similarity between metric synthetic pseudo-features is calculated as follows:

$$\mathcal{L}_{SM} = -log \frac{\exp(sim(\mathfrak{v}_{ij}, \tilde{\mathfrak{v}}_{ji}))}{\sum_{i=1}^{n_s} \sum_{j=1}^{n_q} \exp(sim(\mathfrak{v}_{ij}, \tilde{\mathfrak{v}}_{ji}))}, \tag{12}$$

where $sim(\cdot)$ is the cosine similarity function that measures the synthetic features between the $\mathfrak{v}_{ij}$ and $\tilde{\mathfrak{v}}_{ji}$.

3.6. Full objective of the network

The full objective for training the network consists of three parts: measure support and query sample, measure the synthesis and the original target object feature, and measure synthesis support and query sample pseudo feature, which can be computed as:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{Mb} + \lambda \mathcal{L}_{MGA} + \gamma \mathcal{L}_{GCF} \\ &= \mathcal{L}_{Mb} + \lambda(\mathcal{L}_{CE}^{c} + \alpha \mathcal{L}_{CE}^{f}) + \gamma(\mathcal{L}_{SM} + \mathcal{L}_{MS}), \end{aligned} \tag{13}$$

where parameter α is the trade-off weight for the loss $\mathcal{L}_{CE}^{c}$ Eq. (4) and $\mathcal{L}_{CE}^{f}$ Eq. (5), parameters λ and γ are the weights for the loss $\mathcal{L}_{MGA}$ and $\mathcal{L}_{GCF}$. The settings of these parameters are detailed in the experimental description.

4. Experimental settings

In this section, we first describe six benchmark datasets, including structured and unstructured datasets, for evaluating the proposed method. Then, we give the implementation details of the experimental settings and evaluation metrics.

4.1. Datasets

We utilize the tieredImageNet [23], Fewshot CIFAR100 (FC100) [24], CIFAR-FS [25], miniImageNet [14], CUB-200-2011 (CUB) [26], and Standford-Cars (Cars) [27] datasets extensively employed in current works to ensure a fair evaluation of the proposed MACF performance. We uniformly transform the images resize to 84×84 when the model loads these datasets. We list the primary information of the above datasets in Table 1.

We provide a introduction to the sources of these datasets:

1) Structured datasets: tieredImageNet [23] is a larger subset of the ILSVRC-12 dataset, consisting of 608 classes with a hierarchical data structure. FC100 [24] and CIFAR-FS [25] datasets are all derived from the division of the CIFAR100 dataset. They have a hierarchical data structure, but the data categories are divided differently.
2) Unstructured datasets: miniImageNet [14] is a subset of the ImageNet dataset without the hierarchy of data categories. CUB [26] and Cars [27] are extensive utilized for fine-grained classification, which is a challenge for few-shot learning since the appearance of species is not intrinsically different.

4.2. Implementation details

All our experiments are based on the PyTorch framework and were performed on a Ubuntu20.04 desktop computer, which utilizes NVIDIA GTX3090 with 24.0 GB video memory and a 2.40 GHz × 24 Intel Xeon Silver 4214R CPU. For experiments on ResNet10, ResNet12, and ResNet18, the optimizer used is SGD, and the initial learning rate is set to 0.1. We set the momentum of 0.9 and the weight decay of 0.0005. All our models are end-to-end trained from scratch with no additional dataset or steps. For CIFAR-FS and FC100; miniImageNet and tieredImageNet; CUB and Cars, we follow [13] set the metric module weight to 0.5, 0.25, and 1.5, respectively. Additionally, for CIFAR-FS, FC100, and tieredImageNet datasets with a hierarchical structure, we set the hyper-parameter λ to 4, otherwise to 0. The hyper-parameter γ is set to 0.9 for all datasets. We set the parameter $\alpha = 0.4$. We report the test evaluation experiment 2000 times on the test set and their mean accuracy with 95% confidence intervals.

4.3. Evaluation metrics

Tree-induced error (TIE): In the dataset with hierarchy, we measure the distance between the true and predicted label via TIE, which is defined as follows:

$$TIE(y, y^*) = |E_h(y, y^*)|, \tag{14}$$

where $E_h(y, y^*)$ indicates the set of edges from y to y^*, y and y^* denote the true and predicted label of sample x, respectively.

Hierarchical F-measure evaluation (F_H): We evaluate the multi-grained hierarchy via F_H, which is defined as follows:

$$\begin{cases} F_H(y, y^*) = \dfrac{2 \cdot P_H R_H}{P_H + R_H}; \\ P_H(y, y^*) = \dfrac{|Y \bigcap Y^*|}{|Y|}; \\ R_H(y, y^*) = \dfrac{|Y \bigcap Y^*|}{|Y^*|}, \end{cases} \tag{15}$$

where Y and Y^* indicate the set of the true and predicted class containing coarse nodes, respectively. P_H represents hierarchical precision and R_H means recall rate.

5. Experimental results and analysis

The best results of each set are highlighted in bold. Note that the experimental results from the original author are represented by "*", "-" indicates that the method is not reported, and "⋆" denotes larger backbones than ResNet12.

5.1. Comparison with related methods

We compare our method with metric-learning-based approaches, as shown in Table 2. The table presents a comparison of 5-way classification accuracy (%) for relevant methods, along with 95% confidence intervals, on the CIFAR-FS and FC100 datasets. From these results, we can derive the following conclusions:

ProtoNet [15] constructs class prototypes and measures distances between them for classification. MetaOptNet [33] employs meta-learning to optimize linear classifiers, significantly improving high-dimensional embeddings. Our proposed MACF method consistently achieves high performance across most datasets. Notably, it outperforms related models by a significant margin, achieving improvements of 3.10% in the 5-shot setting on the FC100 dataset and 2.84% in the 5-shot setting on the CUB dataset. Unlike ResNet12, both the Cosine Classifier [31] and S2M2 [32] utilize the more complex ResNet34 architecture. Despite this, our MACF method surpasses both 1-shot and 5-shot scenarios. Our multi-granularity awareness framework outperforms the recent single-granularity network PHR [29] in the 5-shot experiment. Compared with recent feature generation methods such as SFG [22], our method does not achieve optimal performance in the 1-shot experiment owing to the relatively small number of pseudo-features generated by our fusion approach. However, in the 5-shot setting, our synthesized features slightly outperform SFG while utilizing a more lightweight cross-fusion framework. In both 1-shot and 5-shot classification tasks, MACF surpasses recent methods MetaTEDL [36] and DKPL [37] on the CIFAR-FS and FC100 datasets by leveraging a multi-scale feature fusion mechanism. Notably, MACF exhibits even greater performance gains on FC100, where cross-level feature availability is limited. Additionally, compared with the suboptimal method MTL [40], MACF improves classification accuracy by 1% in the 1-shot setting and 5.5% in the 5-shot setting.

The results on tieredImageNet are summarized in Table 3. From this table, we can draw the following conclusions. HGNN [18] uses a Graph Neural Network (GNN) for feature extraction, and its network architecture is more complex than ResNet12. Similarly, these PPA [42], wDAE-GNN [43], and CTM [44] methods are more complex than ResNet12. HMRN [41] is closely related to MACF, considering semantic information to construct a multi-granularity hierarchical structure to assist classification, while our method MACF is slightly better in classification performance.

The results for the CUB and miniImageNet datasets are summarized in Table 4. From this table, we can derive the following conclusions regarding the generalization ability of the proposed MACF across different few-shot benchmark datasets. Additionally, to assess the impact of fusion-generated features, we evaluate MACF using only the cross-fusion module. MAP-Net [4] integrates

Table 2
5-way classification accuracy (%) comparison of related method with 95% confidence intervals on CIFAR-FS and FC100.

(a) CIFAR-FS

Method	Backbone	1-shot	5-shot
ATSA* [28]	CNN4	67.15 ± 0.30	81.65 ± 0.30
PHR* [29]	CNN4	78.60 ± 0.60	86.50 ± 0.40
Boosting* [30]	WRN-28-10⋆	73.60 ± 0.30	86.00 ± 0.20
Cosine classifier* [31]	ResNet34⋆	61.49 ± 0.91	82.37 ± 0.67
S2M2* [32]	ResNet34⋆	72.92 ± 0.83	86.55 ± 0.51
ProtoNet* [15]	ResNet12	72.20 ± 0.70	83.50 ± 0.50
MetaOptNet* [33]	ResNet12	72.60 ± 0.70	84.30 ± 0.50
RFS-simple* [34]	ResNet12	71.50 ± 0.80	86.00 ± 0.50
RENet [13]	ResNet12	71.70 ± 0.47	86.32 ± 0.32
ConstellationNet* [35]	ResNet12	75.40 ± 0.20	86.80 ± 0.20
MetaTEDL* [36]	ResNet12	71.30 ± 0.90	85.20 ± 0.60
DKPL* [37]	ResNet12	75.66 ± 0.20	87.70 ± 0.13
SFG* [22]	ResNet12	**82.28 ± 0.56**	87.10 ± 0.38
MACF	ResNet12	76.38 ± 0.48	**87.73 ± 0.31**

(b) FC100

Method	Backbone	1-shot	5-shot
Baseline2020* [38]	ResNet12	36.82 ± 0.51	49.72 ± 0.55
MetaOptNet* [33]	ResNet12	41.10 ± 0.60	55.50 ± 0.60
TADAM* [24]	ResNet12	40.10 ± 0.40	56.10 ± 0.40
RENet [13]	ResNet12	43.63 ± 0.41	59.89 ± 0.40
MixtFSL* [39]	ResNet12	41.50 ± 0.67	58.39 ± 0.62
ConstellationNet* [35]	ResNet12	43.80 ± 0.20	59.70 ± 0.20
MetaTEDL* [36]	ResNet12	44.00 ± 0.80	60.00 ± 0.70
DKPL* [37]	ResNet12	45.60 ± 0.20	60.95 ± 0.13
MTL* [40]	ResNet12	45.30 ± 1.80	57.50 ± 0.80
MACF	ResNet12	**46.36 ± 0.40**	**62.99 ± 0.41**

Table 3
5-way classification accuracy (%) comparison of related method with 95% confidence intervals on tieredImageNet dataset.

Method	Backbone	tieredImageNet	
		1-shot	5-shot
HMRN* [41]	CNN4	57.98 ± 0.26	74.70 ± 0.24
DKPL* [37]	CNN4	56.89 ± 0.22	74.25 ± 0.15
HGNN* [18]	GNN	64.32 ± 0.49	83.34 ± 0.45
PPA* [42]	WRN-28-10⋆	65.65 ± 0.92	83.40 ± 0.65
wDAE-GNN* [43]	WRN-28-10⋆	68.18 ± 0.16	83.09 ± 0.12
CTM* [44]	ResNet18	68.41 ± 0.39	84.28 ± 1.73
MatchNet* [14]	ResNet12	68.50 ± 0.92	80.60 ± 0.71
ProtoNet* [15]	ResNet12	68.23 ± 0.23	84.03 ± 0.16
FEAT* [45]	ResNet12	70.80 ± 0.23	84.79 ± 0.16
RENet [13]	ResNet12	70.79 ± 0.50	84.93 ± 0.35
MTL* [40]	ResNet12	65.30 ± 1.80	81.20 ± 0.80
LCR* [5]	ResNet12	68.58 ± 0.23	84.92 ± 0.16
MACF	ResNet12	**71.92 ± 0.52**	**85.89 ± 0.36**

semantic information with visual features to address spatial asymmetry, significantly enhancing classification performance. In the 1-shot experiment on the CUB dataset, MACF (w/o MGA) exhibits slightly lower performance than MAP-Net. MACF achieves higher accuracy than recent methods CDKT [46] and DKPL [37], potentially owing to its deeper network structure, which enhances representation capabilities. Notably, compared with the ISC [2], which focuses on local regions, MACF improves classification accuracy by 7.32% in the 1-shot experiment and 4.02% in the 5-shot experiment on the CUB dataset. These results demonstrate that the Gaussian cross-fusion of fine-grained features in MACF effectively generates synthetic features, contributing to improved classification accuracy.

5.2. Ablation study with different components

Our method is built on metric learning, in which the metric module $\mathcal{L}_{Mb}$ serves as a fundamental baseline. Additional modules are integrated with $\mathcal{L}_{Mb}$ to assess their contributions to model performance. We conduct ablation experiments for each component,

Table 4
5-way classification accuracy (%) comparison of the SOAT with 95% confidence intervals on CUB and miniImageNet datasets.

(a) CUB

Model	Semantic	Backbone	1-shot	5-shot
HGNN* [18]	N	GNN	69.43 ± 0.49	87.67 ± 0.45
ATSA* [28]	N	CNN4	55.85	66.73
CDKT* [46]	N	CNN4	65.21 ± 0.45	79.10 ± 0.33
DeepEMD* [7]	N	ResNet12	75.65 ± 0.83	88.69 ± 0.50
BPL* [1]	N	ResNet12	-	-
MAP-Net* [4]	Y	ResNet12	**82.45 ± 0.23**	88.30 ± 0.17
LCR* [5]	N	ResNet12	69.57 ± 0.25	81.73 ± 0.16
ISC* [2]	N	ResNet12	73.72 ± 0.69	87.51 ± 0.40
FEAT* [45]	N	ResNet12	73.27 ± 0.22	85.77 ± 0.14
MACF (w/o MGA)	N	ResNet12	81.04 ± 0.43	**91.53 ± 0.17**

(b) miniImageNet

Model	Semantic	Backbone	1-shot	5-shot
HGNN* [18]	N	GNN	60.03 ± 0.51	79.64 ± 0.36
ATSA* [28]	N	CNN4	52.68 ± 0.51	70.91 ± 0.85
DKPL* [37]	N	CNN4	55.28 ± 0.20	71.57 ± 0.16
CDKT* [46]	N	CNN4	47.54 ± 0.21	63.79 ± 0.15
DeepEMD* [7]	N	ResNet12	65.91 ± 0.82	82.41 ± 0.56
BPL* [1]	N	ResNet12	59.57 ± 0.63	76.86 ± 0.49
MAP-Net* [4]	Y	ResNet12	-	-
LCR* [5]	N	ResNet12	63.47 ± 0.20	81.27 ± 0.15
ISC* [2]	N	ResNet12	66.88 ± 0.20	82.22 ± 0.14
FEAT* [45]	N	ResNet12	66.78 ± 0.20	82.05 ± 0.14
MetaTEDL* [36]	N	ResNet12	60.40 ± 0.80	78.00 ± 0.60
MACF (w/o MGA)	N	ResNet12	**67.30 ± 0.23**	**83.31 ± 0.17**

Table 5
Ablation for each part.

$\mathcal{L}_{Mb}$	$\mathcal{L}_{MGA}$	$\mathcal{L}_{GCF}$	CIFAR-FS 1-shot	CIFAR-FS 5-shot
✓			74.384 ± 0.47	86.315 ± 0.32
✓	✓		75.695 ± 0.46	86.937 ± 0.33
✓		✓	75.945 ± 0.48	87.215 ± 0.32
✓	✓	✓	**76.384 ± 0.48**	**87.725 ± 0.31**

and the results are presented in Table 5. With each added component, classification performance improves to varying degrees. These findings indicate that leveraging semantic knowledge to guide the construction of multi-granularity awareness enhances inter-class discrimination. However, in cases of data scarcity, this approach may lead to overfitting. To mitigate this issue, the cross-fusion module of our model, guided by multi-granularity knowledge, effectively addresses both indistinct inter-class boundaries and intra-class space constraints.

Fig. 5 illustrates the impact of different parameter weights on fine-grained classification. In this analysis, our method excludes the multi-granularity perception module and relies solely on the Gaussian cross-fusion (GCF) module. In the 1-shot scenario, the participation weight of the GCF module has a more significant effect on classification performance. Conversely, in the 5-shot setting, the influence of the participation weight of the GCF module remains relatively stable.

5.3. Ablation study with different component parameters

We explore different degrees of the module of participation affect classification. The experimental on CIFAR-FS (benchmark data with hierarchical information) is listed in Table 6, and the analysis mainly has the following two points:

1) We fixed the weights of loss $\mathcal{L}_{Mb} \to 1$ and $\mathcal{L}_{GCF} \to 1$ in the 1-shot and 5-shot experiments on the CIFAR-FS dataset. By adjusting the weight of $\mathcal{L}_{MGA}$, we notice that reasonable participation of coarse and fine-grained is helpful for classification. The experimental results show that when $\lambda = 0.5$ compared with Baseline (BS, only the metric module), MACF improves by 2.2% in the 1-shot experiment. Similarly, as shown in Figs. 6 (a) and (b), when $\lambda = 0.5$, the value of TIE is the smallest, and the value of FH is the largest. The experimental results indicate that when the parameter $\lambda = 0.5$, it is beneficial to construct a multi-granularity structure-assisted classification.

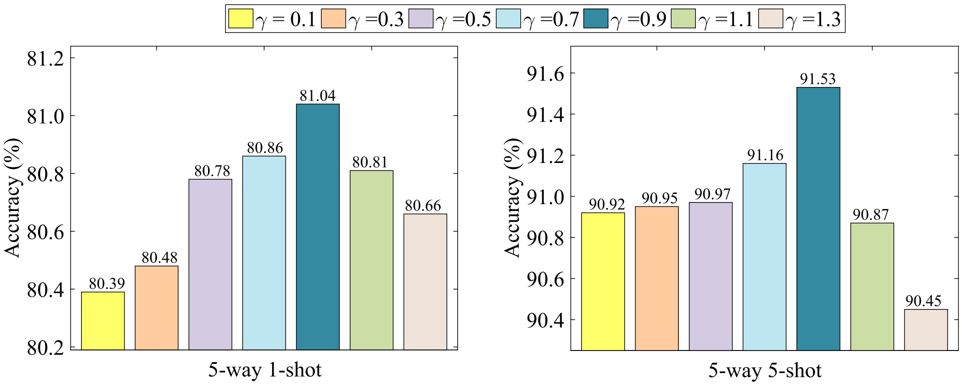

Fig. 5. 5-way classification performance on the CUB dataset without $\mathcal{L}_{MGA}$ by adjusting the weights in the $\mathcal{L}$.

Table 6
5-way classification performance of the proposed MACF by adjusting the weights in the loss function.

λ	γ	CIFAR-FS	
		1-shot	5-shot
1.0	0.9	75.437 ± 0.42	**87.683 ± 0.33**
1.0	0.7	75.061 ± 0.42	87.383 ± 0.32
1.0	0.5	75.473 ± 0.42	87.459 ± 0.32
1.0	0.3	75.742 ± 0.42	87.543 ± 0.33
1.0	0.1	75.891 ± 0.42	87.321 ± 0.33
0.9	1.0	75.415 ± 0.48	87.040 ± 0.32
0.7	1.0	75.750 ± 0.48	87.099 ± 0.33
0.5	1.0	**76.184 ± 0.48**	87.185 ± 0.33
0.3	1.0	75.581 ± 0.49	86.854 ± 0.34
0.1	1.0	71.025 ± 0.50	85.212 ± 0.34

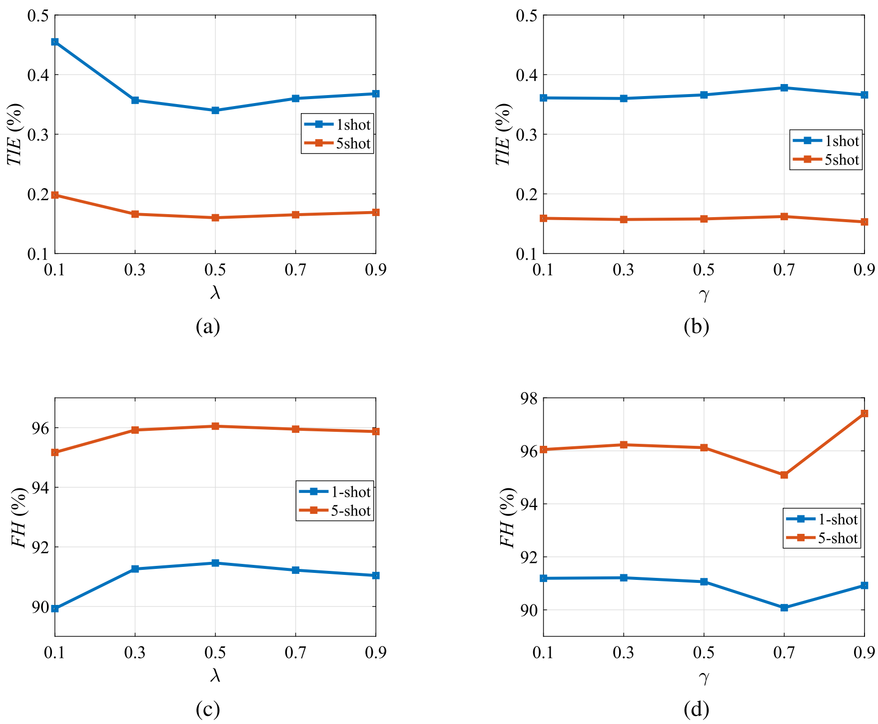

Fig. 6. 5-way classification performance on the CIFAR-FS dataset by adjusting the weights in the loss function. Subfigures (a) and (c) represent the results of TIE and FH of MACF (w/o GCF), respectively. Subfigures (b) and (d) represent the results of TIE and FH of MACF (w/o MGA), respectively.

Table 7
5-way few-shot classification via changing the backbone network on CUB and Cars.

Backbone	Model	1-shot	5-shot
(a) CUB			
ResNet-10	Relation Network† [16]	69.95 ± 0.97	81.67 ± 0.58
	BSNet† [11]	69.73 ± 0.97	82.85 ± 0.61
	RENet [13]	77.32 ± 0.44	89.43 ± 0.27
	MACF (w/o MGA)	**78.11 ± 0.45**	**90.30 ± 0.25**
ResNet-18	Relation Network† [16]	69.58 ± 0.97	83.04 ± 0.60
	BSNet† [11]	69.61 ± 0.92	83.24 ± 0.60
	RENet [13]	78.18 ± 0.14	89.63 ± 0.42
	MACF (w/o MGA)	**78.74 ± 0.45**	**91.01 ± 0.24**
ResNet-12	RENet [13]	78.83 ± 0.43	90.50 ± 0.25
	MACF (w/o MGA)	**81.04 ± 0.43**	**91.53 ± 0.17**
(b) Cars			
ResNet-10	Relation Network† [16]	63.55 ± 1.04	83.85 ± 0.64
	BSNet† [11]	66.97 ± 0.99	84.09 ± 0.66
	RENet [13]	69.84 ± 0.46	**86.12 ± 0.27**
	MACF (w/o MGA)	**69.99 ± 0.47**	85.35 ± 0.29
ResNet-18	Relation Network† [16]	61.02 ± 1.04	83.61 ± 0.68
	BSNet† [11]	60.36 ± 0.98	85.28 ± 0.64
	RENet [13]	**72.17 ± 0.45**	86.30 ± 0.28
	MACF (w/o MGA)	72.03 ± 0.45	**86.34 ± 0.27**
ResNet-12	RENet [13]	72.65 ± 0.46	87.11 ± 0.27
	MACF (w/o MGA)	**72.68 ± 0.45**	**87.40 ± 0.26**

2) Furthermore, we fixed the weights of loss $\mathcal{L}_{Mb} \to 1$ and $\mathcal{L}_{MGA} \to 1$ in the 1-shot and 5-shot experiments on the CIFAR-FS dataset. By adjusting the weight of $\mathcal{L}_{GCF}$, we learned that reasonable participation of cross-fusion is helpful for classification. The experimental results show that MACF improves by 1.3% in the 5-shot experiment when $\gamma = 0.9$ compared with BS. Similarly, when $\gamma = 0.9$, the value of TIE is the smallest, and the value of FH is the largest, as shown in Figs. 6 (c) and (d). The experimental results demonstrate that when the parameter $\gamma = 0.9$, the synthetic features generated by the fusion module are optimal for classification.

5.4. Ablation study with different backbone

To further explore the effectiveness of the proposed method MACF, we change the feature embedding modules to ResNet10 and ResNet18 networks. The experimental results are listed in Table 7. GCF (MACF w/o MGA) represents MACF without using the multi-granularity perception module. GCF is slightly worse than RENet [13] on the 5-way 5-shot classification of the cars dataset when all methods use ResNet10 as the feature embedding module. Similarly, our proposed method MACF (w/o MGA), is closely RENet in the 5-way 1-shot on the cars dataset classification when utilizing the ResNet18 network as the feature embedding module. The experimental results demonstrate that our proposed method MACF (w/o MGA) outperforms the Relation Network [16], BSNet [11], RENet [13] methods in the case of mostly replacing the embedded feature modules. GCF module is a plug-and-play solution that can be embedded in other methods.

5.5. Feature visualization

Additionally, we conduct 5-way 1-shot visualization experiments on the CIFAR-FS and tieredImageNet datasets, randomly selecting test tasks for evaluation. We evaluate three models: Baseline (BS), MGA, and MACF. To visualize their results, we employ t-SNE and Grad-CAM [47], as shown in Figs. 7 and 8. From these visualizations, we can draw the following conclusions:

1) MACF (w/o GCF) enhances inter-class distinction, making class boundaries more distinct. Additionally, incorporating the GCF module further compacts the intra-class feature space, improving feature consistency.
2) MACF focuses more precisely on the region of the target object, leading to a more concentrated and discriminative representation in the classification process. In brief, the MACF method promotes the inter-classes obvious distinction and enriches the diversity of the intra-classes space. Therefore, MACF makes the intra-class space more compact, improving classification accuracy.

The experimental results demonstrate the effectiveness of the MACF method. Although MACF enhances intra-class diversity through cross-fusion, it still requires improvements in generating diverse feature representations, particularly when handling highly similar categories. Additionally, during the feature synthesis process, the sensitivity of the model to outliers or noise may affect its stability and robustness when applied to real-world data.

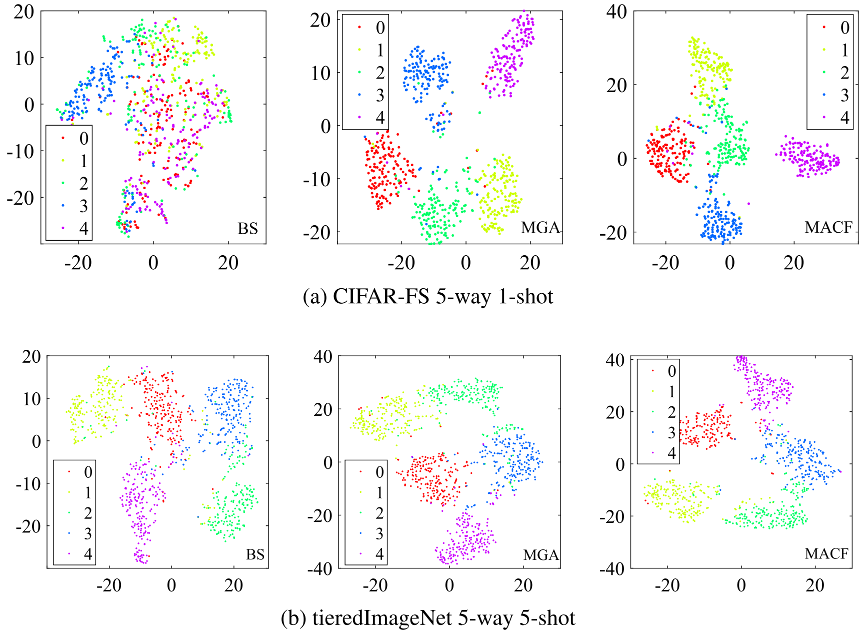

Fig. 7. Illustration of feature visualization from BS, MGA, and MACF on the CIFAR-FS and tieredImageNet datasets. The feature points of different colors represent different classes.

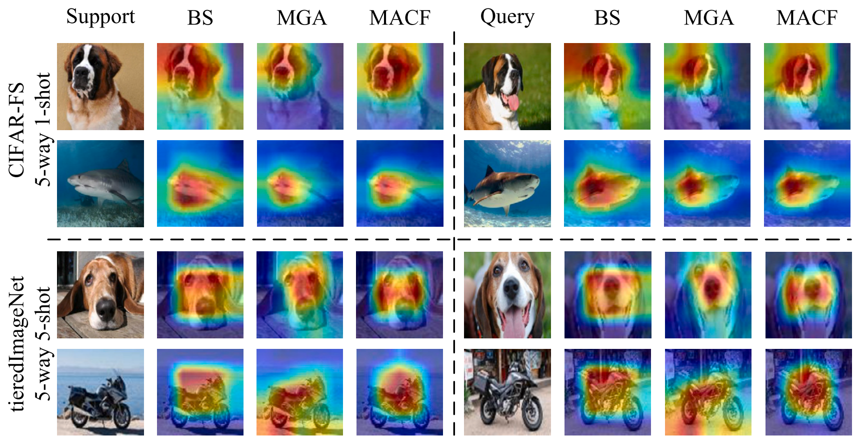

Fig. 8. Illustration of feature heatmaps visualization under BS, MGA, and MACF on the CIFAR-FS and tieredImageNet datasets.

6. Conclusions and future work

In this work, we propose an MACF method for FSL to enhance inter-class separation while enriching intra-class diversity. Specifically, the multi-granularity awareness module facilitates the generation of high-quality fine-grained features, which play a crucial role in improving few-shot classification performance. The cross-fusion module synthesizes features to expand the intra-class feature space and mitigate the fragmented distribution of intra-class representations. Additionally, we introduce a fusion metric to assess the impact of synthetic features, helping to minimize the influence of outliers on the model. MACF achieves superior performance across six few-shot benchmark datasets.

In summary, while MACF has demonstrated significant potential, its performance is highly dependent on the quality and diversity of features extracted from support and query samples. When the available data are highly homogeneous or lack sufficient variability, the effectiveness of this approach may be compromised. To address these limitations, we plan to develop strategies that enhance feature diversity in support and query samples. This could involve incorporating self-paced hard task-example mining [48] to prioritize more challenging samples or leveraging attention mechanisms [49] to focus on the most informative regions of the data. By implementing these improvements, we aim to mitigate existing challenges and enhance the robustness of our approach in future work.

CRediT authorship contribution statement

Zhiping Wu: Writing – original draft. **Dongqing Li:** Writing – review & editing, Data curation. **Linhua Zou:** Writing – review & editing, Methodology. **Hong Zhao:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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